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Via email to: AirCleaners2021STD0035@ee.doe.gov

Dr. Stephanie Johnson
Office of Energy Efficiency and Renewable Energy
Building Technologies Office, EE-5B
1000 Independence Avenue SW
Washington, DC 20585

Docket Number: EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036

Subject: Response to Air Cleaner Standards and Test Procedures Request for Information

Dear Dr. Johnson:

Trane Technologies welcomes the opportunity to comment on the Request for Information on air cleaner standards and test procedures.

Trane Technologies ("Trane") is a climate company with well-known brands such as Trane and Thermo King, global leaders in building and transport air conditioning and transport refrigeration products. Trane is well known for global leadership in decarbonizing its own footprint as well as customers' buildings and transport applications to reduce climate change. Trane's Gigaton Challenge is a commitment to reduce its customer emissions by 1 billion metric tons by 2030.¹ Success primarily requires the decarbonization of new and existing buildings. Trane began commercializing low carbon products in 2014 and has been trying to reduce energy consumption in buildings for decades. Unfortunately, the current pandemic response has been largely without regard to energy efficiency, which can significantly increase the use of energy and carbon emissions from buildings.

The requested information is discussed below:

¹ Trane Technologies, 2020. "Bold Action for a Sustainable Future: 2020 ESG Report."
<https://www.tranetechnologies.com/en/index/sustainability/sustainability-reports/esg-report.html>

1. ***The DOE requests information regarding the development and evaluation of potential new energy conservation standards for consumer air cleaners, and whether such standards would result in significant energy savings, be technologically feasible and economically justified.***

The establishment of a new energy conservation standard is necessary for consumer air cleaners, especially portable or in-room filter-based units, because consumers need guidance and the opportunity to avoid unnecessary energy consumption. Our customers are experiencing unprecedented energy bills as they incorporate portable air cleaners.

Since there are no efficiency standards today for this equipment, a new standard should result in significant energy savings because the less efficient equipment would be prohibited. The new standard is technologically feasible because it could utilize clean air delivery rate per watt, which would eliminate products that are not effective.

2. ***The DOE requests comment on its example approach of defining an integrated clean air delivery rate per watt metric.***

The integrated clean air delivery rate per watt metric approach is appropriate for this standard because it balances product efficacy and energy efficiency. However, additional metrics should be considered as well, such as noise thresholds to avoid occupant space disruption and lack of use. We have seen examples of K-12 classrooms with units that are pushed into a corner and disconnected due to disrupted levels of noise.

3. ***The DOE suggests that technologies such as ionizers, electrostatic precipitators, hydroxyl generators, photocatalytic oxidation, and HEPA filters would be within the scope of the new energy conservation standard.***

Energy conservation standards for equipment are very specific in nature. It would be difficult to include all the suggested technologies within one standard. We suggest prioritizing portable HEPA and other high filter efficiency filter-based units first because of their prevalence in classrooms.

Additionally, some of these technologies can introduce unintended byproducts^{2,3}, which makes a standard more complicated. Specifically, electronic/reactive air cleaning devices may produce byproducts such as ozone, organic acids, and ultrafine particles, which creates the need to require testing to additional standards.

² Siegel JA: Primary and secondary consequences of indoor air cleaners. Indoor Air. 26:88-96, 2016.

³ Sleiman M, Fisk WJ. Evaluation of Pollutant Emissions from Portable Air Cleaners. Final Report: Contract №10–320. Berkeley, CA, December 2014.

4. The DOE requests additional information that should be considered.

As mentioned above, noise can be very disruptive in occupied spaces. Self-contained/portable air cleaners may be the only immediately implementable way to address localized IAQ problems in various building types but to deliver these benefits, the unit must operate. Portable in-room air cleaners may introduce noise that disrupts normal use of the space. This can be particularly problematic in K-12 educational environments and lead educators to turn in-room air cleaners off.⁴

5. The DOE requests comments on whether capacity or any other performance-related features, such as air cleaning technology (i.e., whether the product destroys or deactivates contaminants from the air or simply removes them), of consumer air cleaners would justify the establishment of different product classes.

Users have varying performance goals and technologies vary in efficacy, so different product classes could be useful to consumers. Efficacy based on the targeted solution should drive the differentiation. Other market segmentation could be useful based on use (e.g., K-12, assisted living, wildfire areas, etc.).

6. DOE requests comment on test methods that could be used to test “non-portable” consumer air cleaners, such as those that are permanently mounted (e.g., ceiling-mounted air cleaners) or that provide whole-home air cleaning in conjunction with central heating or air conditioning systems?

Permanently mounted HVAC systems in buildings are already substantially regulated through building codes, product standards, and DOE regulations. No additional standards should be considered without an exhaustive and multi-stakeholder consultative process.

Please reach out for further discussion.

Sincerely,



Nanette Lockwood
Sr. Director, Policy & Advocacy
Trane Technologies

⁴ J. Curtius, M. Granzin & J. Schrod (2021) Testing mobile air purifiers in a school classroom: Reducing the airborne transmission risk for SARS-CoV-2, *Aerosol Science and Technology*, 55:5, 586-599, DOI: [10.1080/02786826.2021.1877257](https://doi.org/10.1080/02786826.2021.1877257)